

$$\left[\begin{array}{cccccccc}
\frac{D_{zz}^{12}+D_{zz}^{13}+D_{zz}^{23}+3J}{4} & -\frac{\sqrt{3}(D_{xz}^{12}+iD_{yz}^{12}+D_{xz}^{13}+iD_{yz}^{13}+D_{xz}^{23}+iD_{yz}^{23})}{6} & \frac{\sqrt{3}(D_{xx}^{12}+2iD_{xy}^{12}-D_{yy}^{12}+D_{xx}^{13}+2iD_{xy}^{13}-D_{yy}^{13}+D_{xx}^{23}+2iD_{xy}^{23}-D_{yy}^{23})}{12} & 0 & -\frac{\sqrt{2}(D_{xz}^{13}+iD_{yz}^{13}-D_{xz}^{23}-iD_{yz}^{23}+2id_x^{12}-2d_y^{12}-id_x^{13}+d_y^{13}-id_x^{23}+d_y^{23})}{8} & \frac{\sqrt{2}(D_{xx}^{13}+2iD_{xy}^{13}-D_{yy}^{13}-D_{xx}^{23}-2iD_{xy}^{23}+D_{yy}^{23})}{8} & -\frac{\sqrt{6}\cdot(2D_{xz}^{12}+2iD_{yz}^{12}-D_{xz}^{13}-iD_{yz}^{13}-D_{xz}^{23}-iD_{yz}^{23}-3id_x^{13}+3d_y^{13}+3id_x^{23}-3d_y^{23})}{24} & \frac{\sqrt{6}\cdot(2D_{xx}^{12}+4iD_{xy}^{12}-2D_{yy}^{12}-D_{xx}^{13}-2iD_{xy}^{13}+D_{yy}^{13}-D_{xx}^{23}-2iD_{xy}^{23}+D_{yy}^{23})}{24} \\
\frac{\sqrt{3}(D_{xz}^{12}-iD_{yz}^{12}+D_{xz}^{13}-iD_{yz}^{13}+D_{xz}^{23}-iD_{yz}^{23})}{6} & \frac{2D_{xx}^{12}+2D_{yy}^{12}-D_{zz}^{12}+2D_{xx}^{13}+2D_{yy}^{13}-D_{zz}^{13}+2D_{xx}^{23}+2D_{yy}^{23}-D_{zz}^{23}+9J}{12} & 0 & \frac{\sqrt{3}(D_{xx}^{12}+2iD_{xy}^{12}-D_{yy}^{12}+D_{xx}^{13}+2iD_{xy}^{13}-D_{yy}^{13}+D_{xx}^{23}+2iD_{xy}^{23}-D_{yy}^{23})}{12} & \frac{\sqrt{6}(D_{xx}^{13}+D_{yy}^{13}-2D_{zz}^{13}-D_{xx}^{23}-D_{yy}^{23}+2D_{zz}^{23}-4id_z^{12}+2id_z^{13}+2id_z^{23})}{24} & \frac{\sqrt{6}\cdot(3D_{xz}^{13}+3iD_{yz}^{13}-3D_{xz}^{23}-3iD_{yz}^{23}-2id_x^{12}+2d_y^{12}+id_x^{13}-d_y^{13}+id_x^{23}-d_y^{23})}{24} & \frac{\sqrt{2}\cdot(2D_{xx}^{12}+2D_{yy}^{12}-4D_{zz}^{12}-D_{xx}^{13}-D_{yy}^{13}+2D_{zz}^{13}-D_{xx}^{23}-D_{yy}^{23}+2D_{zz}^{23}+6id_z^{13}-6id_z^{23})}{24} & \frac{\sqrt{2}\cdot(2D_{xz}^{12}+2iD_{yz}^{12}-D_{xz}^{13}-iD_{yz}^{13}-D_{xz}^{23}-iD_{yz}^{23}+id_x^{13}-d_y^{13}-id_x^{23}+d_y^{23})}{24} \\
\frac{\sqrt{3}(D_{xx}^{12}-2iD_{xy}^{12}-D_{yy}^{12}+D_{xx}^{13}-2iD_{xy}^{13}-D_{yy}^{13}+D_{xx}^{23}-2iD_{xy}^{23}-D_{yy}^{23})}{12} & 0 & \frac{2D_{xx}^{12}+2D_{yy}^{12}-D_{zz}^{12}+2D_{xx}^{13}+2D_{yy}^{13}-D_{zz}^{13}+2D_{xx}^{23}+2D_{yy}^{23}-D_{zz}^{23}+9J}{12} & \frac{\sqrt{3}(D_{xz}^{12}+iD_{yz}^{12}+D_{xz}^{13}+iD_{yz}^{13}+D_{xz}^{23}+iD_{yz}^{23})}{6} & \frac{\sqrt{6}\cdot(3D_{xz}^{13}-3iD_{yz}^{13}-3D_{xz}^{23}+3iD_{yz}^{23}+2id_x^{12}+2d_y^{12}-id_x^{13}-d_y^{13}-id_x^{23}-d_y^{23})}{24} & -\frac{\sqrt{6}(D_{xx}^{13}+D_{yy}^{13}-2D_{zz}^{13}-D_{xx}^{23}-D_{yy}^{23}+2D_{zz}^{23}+4id_z^{12}-2id_z^{13}-2id_z^{23})}{24} & \frac{\sqrt{2}\cdot(2D_{xz}^{12}-2iD_{yz}^{12}-D_{xz}^{13}+iD_{yz}^{13}-D_{xz}^{23}+iD_{yz}^{23}-id_x^{13}-d_y^{13}+id_x^{23}+d_y^{23})}{8} & \frac{\sqrt{2}\cdot(2D_{xx}^{12}+2D_{yy}^{12}-4D_{zz}^{12}-D_{xx}^{13}-D_{yy}^{13}+2D_{zz}^{13}-D_{xx}^{23}-D_{yy}^{23}+2D_{zz}^{23}-6id_z^{13}+6id_z^{23})}{8} \\
-\frac{\sqrt{2}(D_{xz}^{13}-iD_{yz}^{13}-D_{xz}^{23}+iD_{yz}^{23}-2id_x^{12}-2d_y^{12}+id_x^{13}+d_y^{13}+id_x^{23}+d_y^{23})}{0} & \frac{\sqrt{3}(D_{xx}^{12}-2iD_{xy}^{12}-D_{yy}^{12}+D_{xx}^{13}-2iD_{xy}^{13}-D_{yy}^{13}+D_{xx}^{23}-2iD_{xy}^{23}-D_{yy}^{23})}{12} & \frac{\sqrt{3}(D_{xz}^{12}-iD_{yz}^{12}+D_{xz}^{13}-iD_{yz}^{13}+D_{xz}^{23}-iD_{yz}^{23})}{6} & \frac{D_{zz}^{12}+D_{zz}^{13}+D_{zz}^{23}+3J}{4} & -\frac{\sqrt{2}(D_{xx}^{13}-2iD_{xy}^{13}-D_{yy}^{13}-D_{xx}^{23}+2iD_{xy}^{23}+D_{yy}^{23})}{8} & -\frac{\sqrt{2}(D_{xz}^{13}-iD_{yz}^{13}-D_{xz}^{23}+iD_{yz}^{23}-2id_x^{12}-2d_y^{12}+id_x^{13}+d_y^{13}+id_x^{23}+d_y^{23})}{8} & -\frac{\sqrt{6}\cdot(2D_{xx}^{12}-4iD_{xy}^{12}-2D_{yy}^{12}-D_{xx}^{13}+2iD_{xy}^{13}+D_{yy}^{13}-D_{xx}^{23}+2iD_{xy}^{23}+D_{yy}^{23})}{8} & -\frac{\sqrt{6}\cdot(2D_{xz}^{12}-2iD_{yz}^{12}-D_{xz}^{13}+iD_{yz}^{13}-D_{xz}^{23}+iD_{yz}^{23}+3id_x^{13}+3d_y^{13}-3id_x^{23}-3d_y^{23})}{24} \\
\frac{\sqrt{2}(D_{xx}^{13}-2iD_{xy}^{13}-D_{yy}^{13}-D_{xx}^{23}+2iD_{xy}^{23}+D_{yy}^{23})}{8} & \frac{\sqrt{6}\cdot(3D_{xz}^{13}-3iD_{yz}^{13}-3D_{xz}^{23}+3iD_{yz}^{23}+2id_x^{12}+2d_y^{12}-id_x^{13}-d_y^{13}-id_x^{23}-d_y^{23})}{24} & \frac{\sqrt{6}\cdot(3D_{xx}^{13}+3iD_{yz}^{13}-3D_{xx}^{23}-3iD_{yz}^{23}-2id_x^{12}+2d_y^{12}+id_x^{13}-d_y^{13}+id_x^{23}-d_y^{23})}{24} & \frac{\sqrt{2}(D_{xx}^{13}+2iD_{xy}^{13}-D_{yy}^{13}-D_{xx}^{23}-2iD_{xy}^{23}+D_{yy}^{23})}{4} & -\frac{D_{xx}^{12}+D_{yy}^{12}+D_{zz}^{12}+3J}{4} & 0 & -\frac{\sqrt{3}(D_{xx}^{13}+D_{yy}^{13}+D_{zz}^{13}-D_{xx}^{23}-D_{yy}^{23}-D_{zz}^{23}-2id_z^{12}-2id_z^{13}-2id_z^{23})}{12} & -\frac{\sqrt{3}i(d_x^{12}+id_y^{12}+d_x^{13}+id_y^{13}+d_x^{23}+id_y^{23})}{6} \\
\frac{\sqrt{6}\cdot(2D_{xz}^{12}-2iD_{yz}^{12}-D_{xz}^{13}+iD_{yz}^{13}-D_{xz}^{23}+iD_{yz}^{23}+3id_x^{13}+3d_y^{13}-3id_x^{23}-3d_y^{23})}{8} & \frac{\sqrt{2}\cdot(2D_{xx}^{12}+2D_{yy}^{12}-4D_{zz}^{12}-D_{xx}^{13}-D_{yy}^{13}+2D_{zz}^{13}-D_{xx}^{23}-D_{yy}^{23}+2D_{zz}^{23}-6id_z^{13}+6id_z^{23})}{24} & \frac{\sqrt{2}\cdot(2D_{xz}^{12}+2iD_{yz}^{12}-D_{xz}^{13}-iD_{yz}^{13}-D_{xz}^{23}-iD_{yz}^{23}+id_x^{13}-d_y^{13}-id_x^{23}+d_y^{23})}{24} & \frac{\sqrt{2}(D_{xz}^{13}+iD_{yz}^{13}-D_{xz}^{23}-iD_{yz}^{23}+2id_x^{12}-2d_y^{12}-id_x^{13}+d_y^{13}-id_x^{23}+d_y^{23})}{8} & \frac{\sqrt{3}(D_{xx}^{13}+D_{yy}^{13}+D_{zz}^{13}-D_{xx}^{23}-D_{yy}^{23}-D_{zz}^{23}+2id_z^{12}+2id_z^{13}+2id_z^{23})}{8} & \frac{\sqrt{3}i(d_x^{12}+id_y^{12}+d_x^{13}+id_y^{13}+d_x^{23}+id_y^{23})}{4} & \frac{D_{xx}^{12}+D_{yy}^{12}+D_{zz}^{12}-2D_{xx}^{13}-2D_{yy}^{13}-2D_{zz}^{13}-2D_{xx}^{23}-2D_{yy}^{23}-2D_{zz}^{23}-9J}{6} & 0 \\
\frac{\sqrt{6}\cdot(2D_{xx}^{12}-4iD_{xy}^{12}-2D_{yy}^{12}-D_{xx}^{13}+2iD_{xy}^{13}+D_{yy}^{13}-D_{xx}^{23}+2iD_{xy}^{23}+D_{yy}^{23})}{24} & \frac{\sqrt{2}\cdot(2D_{xz}^{12}-2iD_{yz}^{12}-D_{xz}^{13}+iD_{yz}^{13}-D_{xz}^{23}+iD_{yz}^{23}-id_x^{13}-d_y^{13}+id_x^{23}+d_y^{23})}{24} & \frac{\sqrt{2}\cdot(2D_{xx}^{12}+2D_{yy}^{12}-4D_{zz}^{12}-D_{xx}^{13}-D_{yy}^{13}+2D_{zz}^{13}-D_{xx}^{23}+2D_{yy}^{23}-D_{zz}^{23}+6id_z^{13}-6id_z^{23})}{8} & \frac{\sqrt{6}\cdot(2D_{xz}^{12}+2iD_{yz}^{12}-D_{xz}^{13}-iD_{yz}^{13}-D_{xz}^{23}-iD_{yz}^{23}+id_x^{13}-d_y^{13}-id_x^{23}+d_y^{23})}{24} & \frac{\sqrt{3}i(d_x^{12}-id_y^{12}+d_x^{13}-id_y^{13}+d_x^{23}-id_y^{23})}{12} & \frac{\sqrt{3}(D_{xx}^{13}+D_{yy}^{13}+D_{zz}^{13}-D_{xx}^{23}-D_{yy}^{23}-D_{zz}^{23}-2id_z^{12}-2id_z^{13}-2id_z^{23})}{12} & \frac{D_{xx}^{12}+D_{yy}^{12}+D_{zz}^{12}-2D_{xx}^{13}-2D_{yy}^{13}-2D_{zz}^{13}-2D_{xx}^{23}-2D_{yy}^{23}-2D_{zz}^{23}-9J}{12}
\end{array} \right]$$